

LEAN SIX SIGMA — DMAIC PROJECT REPORT

Reducing Customer Checkout Waiting Time in a Retail Clothing Store During Peak Shopping Hours

A Define–Measure–Analyze–Improve–Control (DMAIC) Case Study

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Portfolio Project — Six Sigma Process Improvement

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Methodology: DMAIC | Sector: Retail | Tools: Process Mapping, Fishbone Diagram, 5 Whys, Impact–Effort Matrix

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1. Introduction

Long customer waiting time at checkout is one of the most common causes of dissatisfaction in retail clothing stores, particularly during peak shopping periods such as weekends and holiday seasons. Extended queues not only frustrate customers but also increase pressure on cashiers, raising the likelihood of billing errors and reduced service quality.

This report applies the Six Sigma DMAIC (Define–Measure–Analyze–Improve–Control) methodology to investigate the root causes of long checkout waiting times in a retail clothing store and to propose practical, data-driven improvements. The project is developed independently for educational and portfolio purposes, based on direct observation of a real checkout process and a simulated dataset built from those observations.

The sections that follow walk through each phase of the DMAIC cycle in order: defining the problem and its scope, measuring current performance using key metrics, analyzing the root causes with statistical and visual tools, proposing improvements, and establishing a control plan to sustain the gains achieved.

2. Define Phase

2.1 Problem Statement

During peak shopping hours, particularly on weekends and public holidays between 5:00 PM and 8:00 PM, customers experience excessive waiting times at the checkout counters. Based on direct observation, the average customer waits approximately 12–13 minutes to complete the billing process.

The checkout process is performed by only two billing counters, where each cashier is responsible for scanning items, removing security tags, folding garments, packing purchases, processing payments, and responding to customer queries. Performing multiple tasks simultaneously creates a bottleneck, resulting in longer queues, increased employee stress, and reduced customer satisfaction.

2.2 Observation

The following observations were made during visits to the retail store:

- The store operated with only two checkout counters during peak hours.
- Each cashier performed multiple activities instead of focusing solely on billing.
- Long customer queues formed during busy periods.
- The cashiers appeared rushed and under pressure while serving customers.
- Several customers appeared frustrated due to the extended waiting time.
- The checkout process slowed further when customers purchased multiple clothing items that required folding and packing.

2.3 Objective

The objective of this project is to identify the primary causes of long customer waiting times during peak shopping hours and propose process improvements using the DMAIC (Define–Measure–Analyze–Improve–Control) methodology to reduce checkout waiting time and improve the overall customer experience.

2.4 Project Charter

Project Name: Reducing Customer Checkout Waiting Time During Peak Hours in a Retail Clothing Store Using the DMAIC Methodology

Business Problem: Customers experience long waiting times (approximately 12–13 minutes) during peak shopping hours due to bottlenecks in the checkout process. This leads to customer dissatisfaction, increased employee workload, and reduced operational efficiency.

Objective: Identify the root causes of long checkout waiting times and recommend process improvements using the DMAIC methodology.

Scope

Included in Scope	Excluded from Scope
Checkout process	Inventory management
Queue management	Store layout
Billing activities	Online orders
Packing process	Staffing budget and recruitment
Customer waiting time	Product pricing

Table 1: Project Scope — Included and Excluded Items

Stakeholders: Customers, Cashiers, Store Manager, Sales Associates, Store Management.

Expected Benefits: Reduced customer waiting time; improved customer satisfaction; reduced cashier workload; improved checkout efficiency; better customer flow during peak hours.

Project Timeline: Define and Measure phases completed through direct store observation; Analyze, Improve, and Control phases completed through structured DMAIC analysis of the resulting dataset.

Disclaimer: This project is an independent educational case study based on direct observations of a retail clothing store checkout process. Since access to the retailer's operational data was not available, the dataset used for analysis was simulated using realistic values derived from these observations. The project is intended solely to demonstrate the practical application of the DMAIC methodology and should not be interpreted as an evaluation of any specific retail organization.

3. Measure Phase

3.1 Data Collection

To establish a baseline for the current checkout process, observational data was recorded for 35 customer transactions during peak shopping hours. For each customer, the queue length, waiting time, number of items purchased, billing time, folding and packing time, and number of active counters were logged. The complete dataset is provided in Appendix A.

3.2 Key Performance Indicators

The following key performance indicators (KPIs) were calculated from the observed data to quantify the current state of the checkout process:

KPI	Result	Interpretation
Average Waiting Time	10.29 min	Customers wait more than 10 minutes on average before service begins.
Average Billing Time	3.42 min	The billing transaction itself takes over 3 minutes on average.
Average Folding & Packing Time	1.58 min	Folding and packing add additional service time at the checkout counter.
Maximum Waiting Time	14.78 min	During peak periods, customers may wait nearly 15 minutes.
Minimum Waiting Time	7.00 min	Even the shortest observed waiting time is relatively high.
Standard Deviation	2.05 min	Waiting times vary by approximately 2 minutes, indicating inconsistent service performance.

Table 2: Measure Phase — Key Performance Indicators

3.3 Relationship Between Variables

Scatter analysis was carried out to examine how checkout time is influenced by transaction size and queue length.

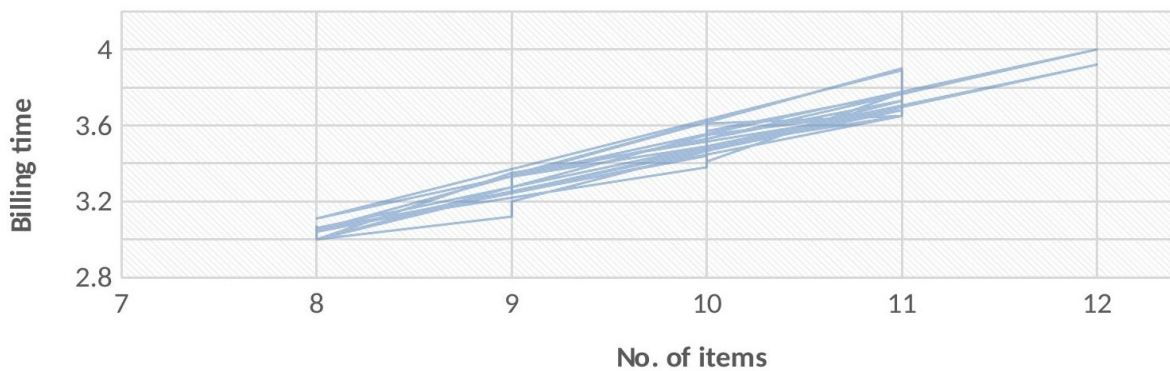


Figure 1: Relationship Between Number of Items and Billing Time

Billing time increases steadily with the number of items purchased, confirming that larger transactions take proportionally longer to scan, tag-remove, and process at the counter.

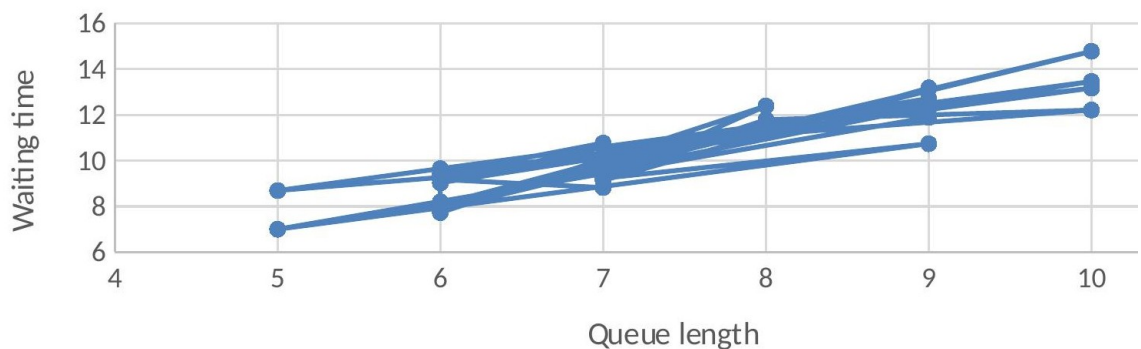


Figure 2: Relationship Between Queue Length and Customer Waiting Time

Customer waiting time rises consistently as queue length increases, showing that with only two active counters, each additional customer in line adds directly to the wait experienced by those behind them.

3.4 Distribution of Customer Waiting Time

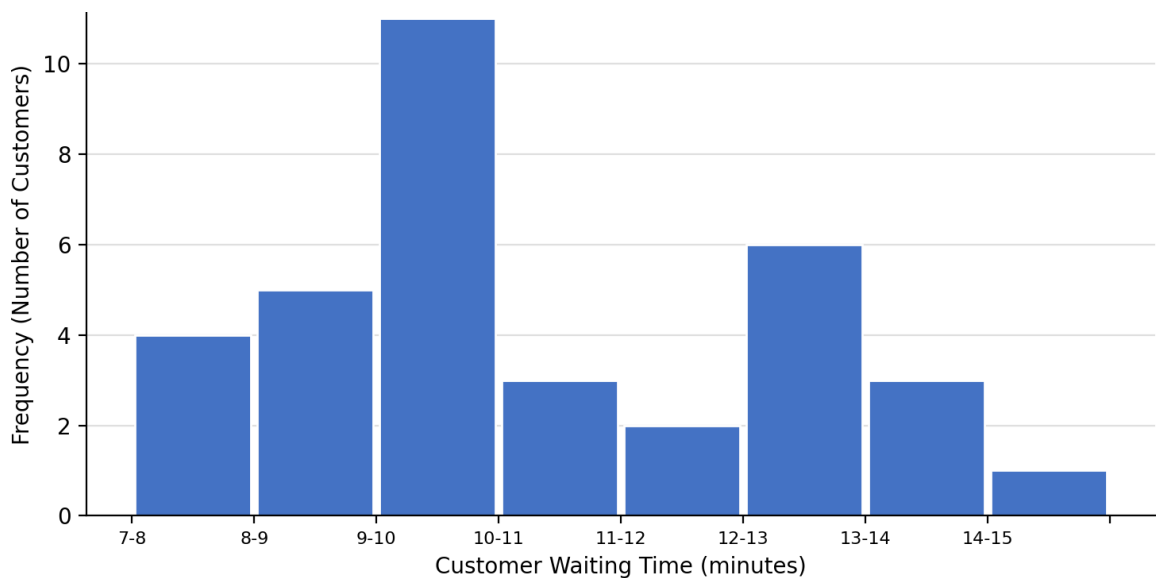


Figure 3: Distribution of Customer Waiting Time

The distribution shows that most customers wait between 9 and 10 minutes, with a secondary cluster in the 12–13 minute range. A right-hand tail extending toward 15 minutes confirms that a meaningful share of customers experience noticeably longer waits during the busiest moments observed.

4. Analyze Phase

4.1 Current Process Map

The process map below illustrates the current checkout workflow observed during peak shopping hours. A single cashier is responsible for scanning items, removing security tags, folding garments, packing purchases, processing payments, and printing receipts. This sequence of multiple activities performed by one employee indicates a potential process bottleneck that contributes to increased customer waiting times.

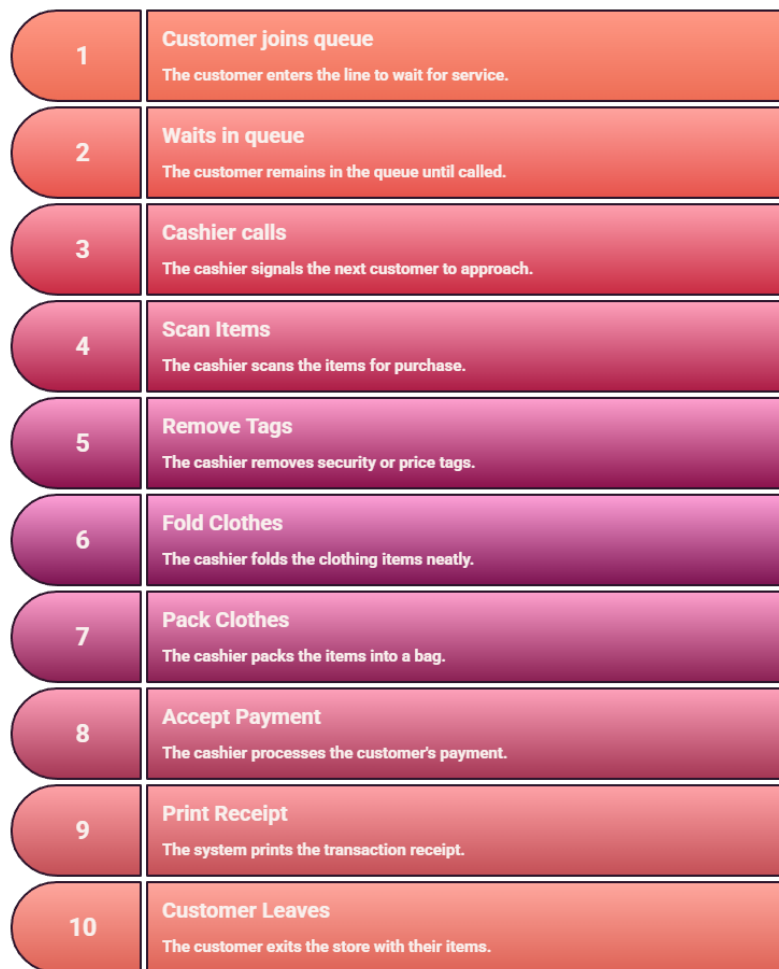


Figure 4: Current Checkout Process Map

4.2 Fishbone (Ishikawa) Diagram

The Fishbone (Ishikawa) diagram was used to systematically identify the potential causes contributing to long customer waiting times during peak shopping hours. The analysis grouped possible causes into five categories: People, Process, Equipment, Management, and Environment. Initial observations indicate that multiple tasks performed by a single cashier, combined with limited checkout capacity during high-demand periods, are the primary contributors to the observed bottleneck.

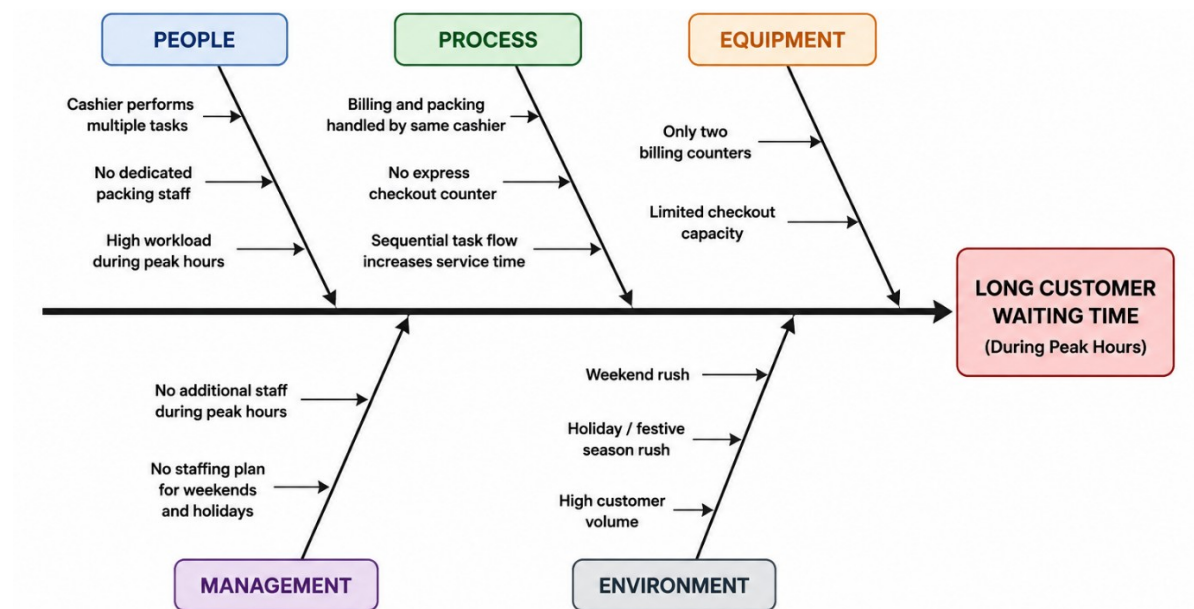


Figure 5: Fishbone (Ishikawa) Diagram — Root Cause Analysis of Long Waiting Time

4.3 5 Whys Analysis

Why Level	Analysis
Problem	Customers experience long waiting times during peak shopping hours.
Why 1?	Because each customer spends a long time at the checkout counter.
Why 2?	Because the cashier performs multiple tasks, including scanning, removing security tags, folding clothes, packing purchases, processing payment, and printing receipts.
Why 3?	Because there is no dedicated packing staff or support during busy periods.
Why 4?	Because the checkout process has not been optimized for peak-hour demand, and only two billing counters are available.
Why 5?	Because staffing and checkout operations are planned similarly for normal and peak business periods, creating a bottleneck during high customer demand.

Table 3: 5 Whys Analysis

4.4 Root Cause

The primary root cause of excessive customer waiting time is an inefficient checkout process in which a single cashier performs multiple sequential tasks without dedicated packing support. This, combined with limited checkout capacity during peak hours, creates a bottleneck that increases customer waiting time.

5. Improve Phase

5.1 Improvement Objective

The objective of the Improve phase is to redesign the checkout process by eliminating identified bottlenecks, reducing customer waiting time, improving cashier efficiency, and enhancing the overall customer checkout experience during peak shopping hours.

5.2 Improvement Opportunities

Issue Identified	Proposed Improvement	Expected Benefit
Cashier performs multiple tasks	Assign one packing assistant during peak hours	Reduces cashier workload
Only two billing counters	Open a temporary third counter during weekends/holidays	Reduces queue length
Long queues	Introduce an express checkout (1–5 items)	Faster service for small purchases
Customers ask questions at checkout	Assign one floor staff member to assist customers	Prevents billing interruptions
Billing and packing performed sequentially	Packing starts immediately after scanning/payment by a dedicated staff member	Shorter checkout cycle time

Table 4: Improvement Opportunities

5.3 Prioritization — Impact-Effort Matrix

The proposed improvements were prioritized using an Impact-Effort matrix to determine which changes should be implemented first.

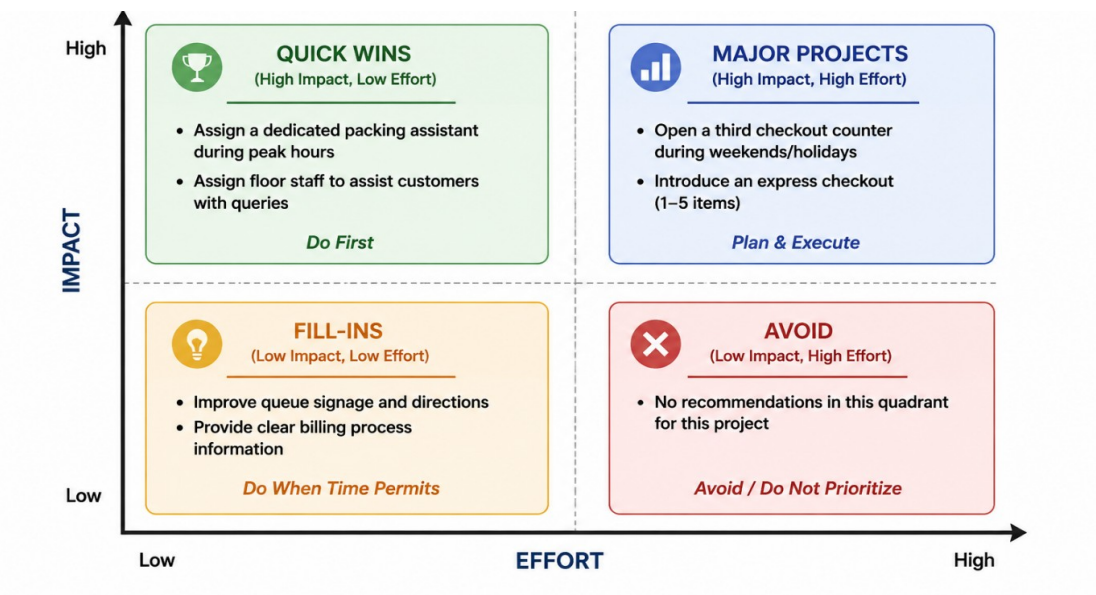


Figure 6: Impact-Effort Matrix — Prioritization of Improvement Opportunities

5.4 Proposed Checkout Process

Figure 7 illustrates the proposed checkout process after implementing the recommended improvements. By assigning folding and packing activities to a dedicated packing assistant, the cashier can focus primarily on billing and payment processing. This change is expected to reduce checkout cycle time, improve cashier productivity, and decrease customer waiting time during peak shopping hours.

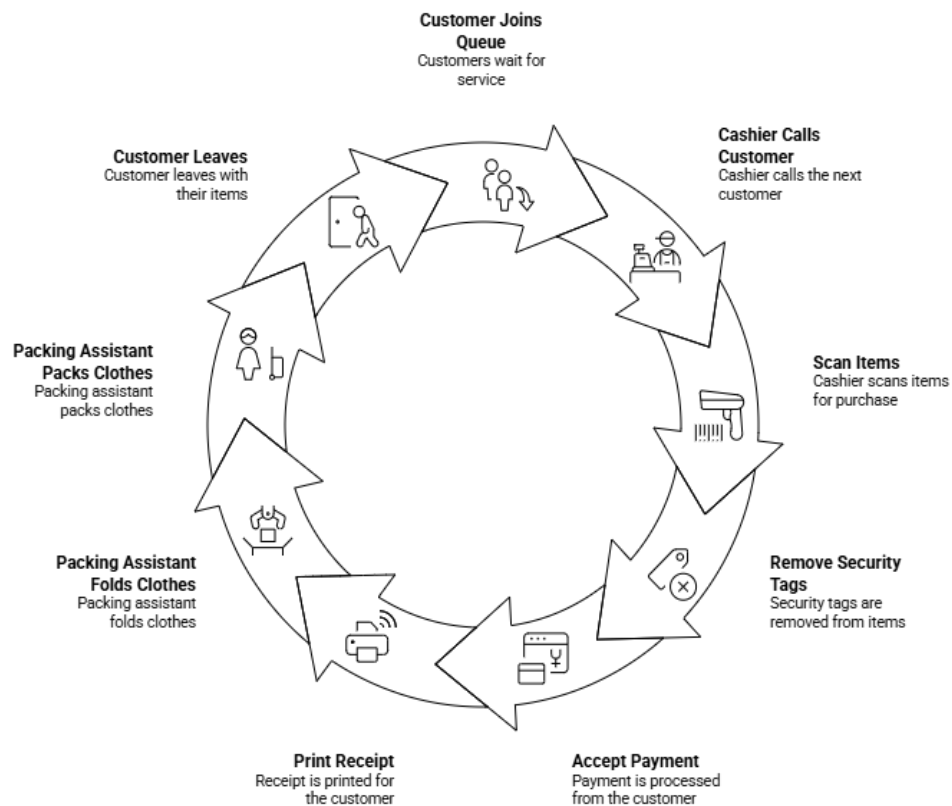


Figure 7: Proposed Checkout Process Cycle (After Improvement)

5.5 Expected Impact on Key Performance Indicators

KPI	Current	Expected
Average Waiting Time	10.29 min	6–7 min
Billing Time	3.42 min	2.5–3 min
Cashier Workload	High	Moderate
Customer Satisfaction	Moderate	High

Table 5: Key Performance Indicators — Current vs. Expected

5.6 Improvement Summary

Based on the analysis conducted using the DMAIC methodology, the proposed improvements focus on reducing cashier workload, improving checkout efficiency, and minimizing customer waiting time. Assigning a dedicated packing assistant during peak hours, opening an additional checkout counter when demand is high, and introducing an express checkout lane are expected to significantly improve service efficiency and enhance the customer experience.

6. Control Phase

6.1 Control Objective

The objective of the Control phase is to ensure that the proposed improvements are maintained over time by continuously monitoring checkout performance, employee workload, and customer waiting time during peak shopping hours.

6.2 Control Plan

KPI	Target	Monitoring Frequency	Responsible Person	Corrective Action
Average Waiting Time	< 7 min	Weekly	Store Manager	Review staffing during peak hours
Billing Time	< 3 min	Weekly	Checkout Supervisor	Observe cashier performance and provide coaching
Queue Length	< 5 customers	Daily (Peak Hours)	Shift Supervisor	Open an additional billing counter if required
Customer Satisfaction	> 90%	Monthly	Store Manager	Review customer feedback and implement improvements

Table 6: Control Plan

6.3 Standard Operating Procedure (SOP)

1. Assign one dedicated packing assistant during weekends and holidays.
2. Open an additional checkout counter whenever customer queues exceed five people.
3. Use an express checkout lane for customers purchasing five items or fewer.
4. Ensure floor staff assist customers with product queries instead of interrupting cashiers.
5. Review checkout performance weekly and take corrective action if KPIs are not achieved.

6.4 Control Summary

The proposed control measures ensure that the improvements achieved through the DMAIC methodology are sustained over time. Continuous monitoring of key performance indicators and adherence to standard operating procedures will help maintain efficient checkout operations and improve customer satisfaction.

7. Conclusion

This project applied the DMAIC (Define–Measure–Analyze–Improve–Control) methodology to evaluate the checkout process of a retail clothing store during peak shopping hours. The study was based on direct observations of the checkout process, supported by a simulated dataset developed for educational purposes.

During the Define phase, the primary problem identified was excessive customer waiting time caused by bottlenecks in the checkout process. The Measure phase established the baseline performance using key performance indicators, including average waiting time, billing time, and folding and packing time. The Analyze phase employed statistical charts, a process map, a Fishbone (Ishikawa) diagram, and the 5 Whys technique to identify the major factors contributing to long queues. The analysis indicated that the checkout process is slowed because a single cashier performs multiple sequential tasks while only two billing counters are available during peak shopping periods.

Based on these findings, several improvements were proposed, including assigning a dedicated packing assistant during busy hours, opening an additional checkout counter when demand increases, introducing an express checkout lane for customers with fewer items, and assigning floor staff to handle customer queries. These recommendations are expected to reduce cashier workload, improve operational efficiency, and enhance the overall customer experience.

Finally, a control plan was developed to ensure that the proposed improvements can be sustained through continuous monitoring of key performance indicators, regular performance reviews, and standardized operating procedures. Although the recommendations were not implemented in this study, the project demonstrates how the DMAIC methodology can be effectively applied to identify process bottlenecks and develop practical, data-driven solutions for process improvement in a retail environment.

This project enhanced my understanding of Lean Six Sigma principles and strengthened my ability to apply structured problem-solving, process analysis, root cause identification, and continuous improvement techniques to real-world operational challenges.

Acknowledgment: *This project was independently developed for educational and portfolio purposes. AI assistance (ChatGPT) was used to support report writing, project structuring, and documentation. All process observations, analysis decisions, and improvement recommendations were reviewed and finalized by the author.*

Appendix A: Checkout Observation Data

The table below lists the complete set of 35 customer observations recorded for this study, as compiled in the accompanying workbook (Retail_Checkout_Data.xlsx).

Customer No.	Queue Length	Waiting Time (min)	No. of Items	Billing Time (min)	Folding & Packing (min)	Active Counters
1	5	7.00	8	3.07	1.00	2
2	9	10.74	8	3.00	1.00	2
3	7	9.23	9	3.25	1.21	2
4	8	12.39	12	3.92	2.43	2
5	6	7.73	8	3.06	1.00	2
6	6	9.27	10	3.38	1.58	2
7	5	8.69	10	3.57	1.71	2
8	6	9.65	11	3.77	2.15	2
9	10	13.46	10	3.41	1.57	2
10	6	9.03	10	3.61	1.81	2
11	7	10.61	11	3.65	2.01	2
12	10	12.21	8	3.04	1.01	2
13	8	11.79	11	3.68	2.18	2
14	7	9.45	9	3.35	1.43	2
15	7	9.78	8	3.00	1.00	2
16	7	10.77	10	3.55	1.67	2
17	6	9.41	11	3.68	1.97	2
18	10	13.17	10	3.48	1.73	2
19	9	12.36	9	3.25	1.43	2
20	6	8.10	9	3.34	1.33	2
21	10	14.78	11	3.90	2.06	2
22	6	7.86	8	3.05	1.01	2
23	6	8.20	10	3.44	1.60	2
24	9	11.89	10	3.58	1.70	2

Customer No.	Queue Length	Waiting Time (min)	No. of Items	Billing Time (min)	Folding & Packing (min)	Active Counters
25	9	12.72	10	3.53	1.60	2
26	9	13.18	12	4.00	2.43	2
27	7	9.13	8	3.11	1.10	2
28	7	8.82	8	3.11	1.13	2
29	6	9.18	11	3.89	2.06	2
30	9	12.72	11	3.65	2.06	2
31	9	12.29	9	3.33	1.41	2
32	7	9.98	11	3.73	2.06	2
33	6	8.08	9	3.20	1.38	2
34	5	7.00	9	3.12	1.39	2
35	7	9.44	8	3.00	1.00	2

Table 7: Sample of Checkout Observation Data (Appendix A)